

Jaden Vichob

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EDUCATION

University of California, San Diego

Electrical Engineering B.S.

GPA: 3.44 / 4.00

Sep 2024 – Present

Expected Jun 2027

Relevant coursework:

- Components and Circuits Lab
- Electronic Circuits and Systems
- Fundamentals of Devices and Materials
- Physical and Fourier Optics
- Introduction to Deep Learning
- Advanced Digital Design Project
- Electronic Materials Science of Integrated Circuits
- Differential Equations
- Linear Algebra
- Electromagnetism
- Quantum Physics
- Probability & Statistics

NOTABLE PROJECTS

Optical Imaging & Fourier Optics Lab Portfolio

Mar 2026

- Designed and characterized optical imaging systems using PCX lenses and achromats, measuring resolution with USAF targets and comparing against theoretical Rayleigh limits
- Investigated Fourier optics by analyzing relationships between sample plane and back focal plane (BFP), applying spatial filtering and frequency-domain masking
- Developed Zemax simulations (singlet, Cooke triplet, and zoom lenses), optimizing RMS spot size and evaluating aberration reduction across designs

ECE 111 Term Project - Viterbi Decoder

Mar 2025

- Designed and implemented a Viterbi decoder in SystemVerilog for error correction in communication systems.
- Implemented ACS, BMC, and TBU modules in SystemVerilog for a Viterbi decoder, enabling error correction under noisy channel conditions
- Extensively tested robustness of design by injecting various quantities of inverted bits at varied intervals, deriving a general formula for which the decoder is accurate.

UCSD IEEE Winter 2025 Quarterly Project

Mar 2025

- Participated in a collaborative group to create a portable, solar-powered basketball hoop.
- Acted as lead circuit designer for an Arduino-powered digital counter, using an ultrasonic sensor.
- Achieved 2nd place out of 20 competing teams for functionality, innovation, and presentation.

LEADERSHIP & TEACHING EXPERIENCE

Kung Fu and Chinese Lion Dance Assistant Instructor

2021 - Present

Camarillo Kung Fu and Lion Dance Club

- Instructed and mentored newer students in both weekly group and private practice sessions.
- Occasionally led kung fu classes, in charge of teaching lion dance choreography for public performances, cultural events, and private receptions.
- Important voice in determining adaptations to environmental challenges for lion dance shows.

HKN Outreach Tutor

Mar 2026 – Present

UCSD

- Created and delivered an interactive lesson to present to socioeconomically disadvantaged K-12 students in the San Diego area
- Aimed to inspire and introduce students to engineering principles while fostering a passion for learning and creating
- Designed a lesson teaching basic electrical engineering fundamentals to create a breadboard-based remote-controlled, battery-powered car

RELEVANT SKILLS

- **Hardware & Design:** PCB (KiCAD), Arduino, breadboarding, soldering, oscilloscope
- **Programming:** C, C++, Python, NumPy, PyTorch, MATLAB, Verilog, SystemVerilog
- **Tools:** AMD Vivado, Intel Quartus Prime, ModelSim, LTspice, PSpice, Ansys Zemax, ImageJ
- **CAD:** Fusion360, Onshape
- **Microsoft Office:** Word, Excel, Powerpoint